

WASP-43b

R-0494

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_230331 · created 2026-07-20T11:55:27+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460035.5167 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.183 +/- 0.056
Transit depth (Rp/Rs)^2	3.35 +/- 2.04 [%]
Orbital Inclination (inc)	80.96 +/- 2.89
Airmass coefficient 1 (a1)	1.005 +/- 0.017
Airmass coefficient 2 (a2)	-0.004 +/- 0.036
Scatter in the residuals of the lightcurve fit is	1.73 %
Best Comparison Star	#3 - [462, 249]
Optimal Method	PSF photometry
Transit Duration (day)	0.0 +/- 0.0

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.183 ± 0.056 vs 0.1594 (deviation 0.42σ)
Duration fit vs published	0.0 d vs 0.0512 d (deviation infσ)
Mid-time O-C	1173.8 min
Depth SNR	3.3
Residual scatter	1.73 %

Light curve

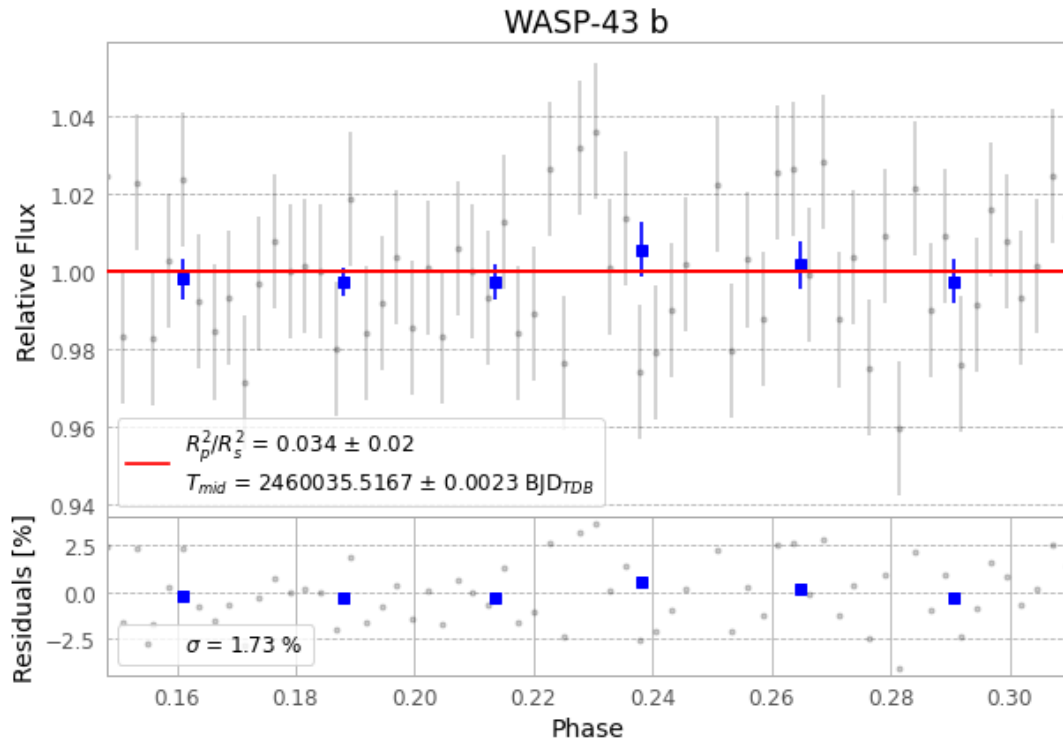


Figure 1 — FinalLightCurve_WASP-43 b_2023-03-31.png (SHA-256 2fd1bb26fe9b6176...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [440, 122], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4c92ec3d8a98a49d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

64 FITS frames (42 MB), WASP-43230401030915.FITS → WASP-43230401061815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-230331.log (d1b0f98c5aec...), FinalLightCurve_WASP-43 b_2023-03-31.png (2fd1bb26fe9b...), FinalLightCurve_WASP-43 b_2023-03-31.csv (338458347425...)

Compute resources

Wall time 80.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 11:55Z.